MYCELIUM · PORTFOLIO · 2024—∞

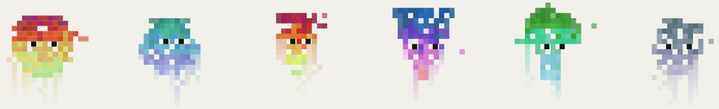
Try it Research ▸ Case study ▸ Design system ▸

A LIVING ARCHIVE OF ANONYMOUS FEELING · 低语菌网

Mycelium

whisper a sentence — a creature grows from how it feels

An anonymous whisper is read by a language model and grown into a one-of-a-kind pixel **spore**, then let loose in a shared colony where everyone's feelings meet and slowly rub off on each other. The same sentence always grows the same spore.



ember drift TENDER · calm bell CALM · sunny crumb CURIOUS · moony moth DREAMY · mint pip COMPANION · quiet wisp LONELY

PROBLEM

Young adults in transition are easy to reach online but often feel disconnected. Platforms surface content, not how a group actually feels.

APPROACH

Turn anonymous disclosure into a living ecology: input → AI reading → morphology → social simulation → shared field.

CONTRIBUTION

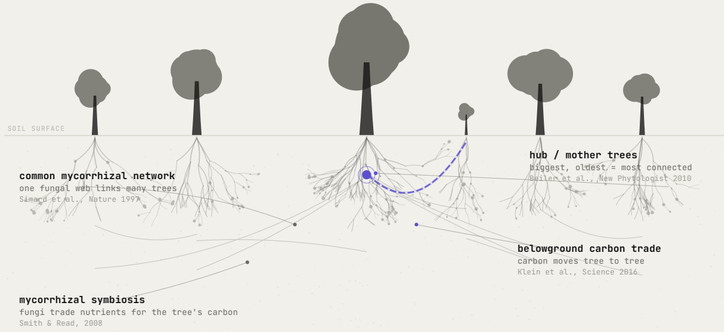
A way to feel present with others without talking — connection shown as colour exchange, care built in as an ecological rule, identity fixed as deterministic form.

ROLE & STATUS

Concept, design & full-stack build · deployed web prototype · formative evaluation in progress.

Why “Mycelium”

The whole thing borrows from **mycorrhizal networks** — the underground web of fungal threads (the *mycelium*) that links the trees of a forest, sometimes called the **“wood-wide web.”** Through it, trees quietly trade sugar and nutrients, pass chemical signals, and older **mother trees** keep seedlings alive. It's connection that is *slow, shared, and out of sight* — the opposite of a feed. Below is the real network it's named after — every claim cited:



Real mycorrhizal science, cited: Simard et al., *Nature* 388:579 (1997) · Klein, Siegwolf & Körner, *Science* 352:342 (2016) · Beiler et al., *New Phytologist* 185:543 (2010) · Smith & Read, *Mycorrhizal Symbiosis* (2008).

Read the forest as a community and the biology stops being about trees. A **community** is a place where what you have is quietly shared — the surplus of one becomes another's survival, the oldest hold up the youngest, and none of it has to be announced to count.

IN A FOREST	IN A COMMUNITY
fungi trade nutrients for the tree's carbon	members sustain each other — no score kept
carbon flows from surplus to deficit	the ones who have more carry the ones who don't
mother trees keep the seedlings alive	you're held by the ones who came before
slow, shared, out of sight	belonging is felt, never performed

That feeling — being quietly held by a group you never had to perform for — is exactly what loneliness is the absence of. It's what **Mycelium** tries to hand back: a place to be present with others without a single word of explanation.

01 Try it — grow a spore right here

MYCELIUM · TREE HOLE 树洞 — TYPE ANYTHING, IN ANY LANGUAGE

today, my heart feels like a quiet lake after rain

grow ▸



Misty Pudding

SPECIMEN

emotion	calm · 平静
intensity	0.79
hue	H 220°
determinism	same sentence → same spore

This preview reads your sentence into one of the **six families** with a lightweight offline heuristic (keyword + hash) and shows that family's **real specimen** — the same pixels the live colony grows. In the deployed product, **Claude** reads the sentence into 18 emotion labels, intensity and six morphology parameters, and grows the exact, unique creature. Grow your own on the live site ▸

02 What it is

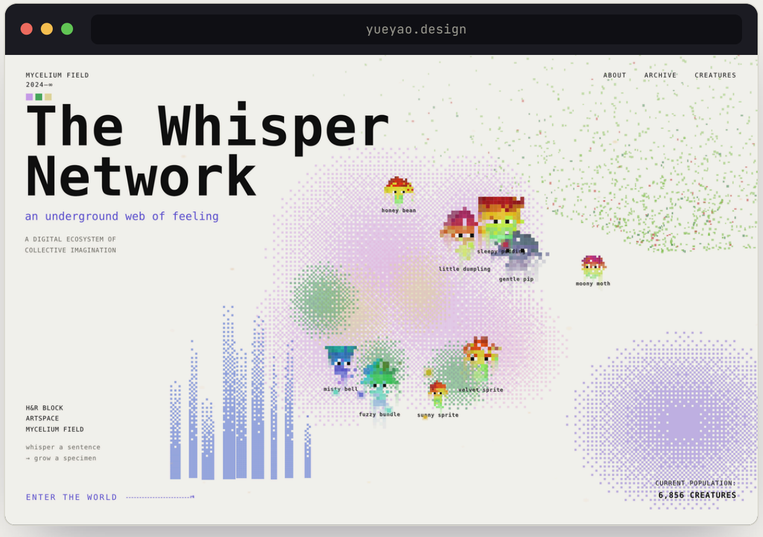
It isn't a mood diary, an AI companion, or a feed. It's a small shared world where private feelings stick around as living creatures.

Everything here is procedural — no drawn sprites, no image model. The reading of a sentence seeds a deterministic random stream, and that stream decides the creature's outline, palette, sharpness and behaviour. So an anonymous feeling turns into something you can **name, claim, share and come back to**, without ever attaching who you are.

whisper	read	grow	meet
one sentence anonymous · text or voice	AI interprets emotion labels · intensity · morphology	a pixel spore deterministic · unique	the colony bonds shown as colour exchange
stay it accumulates a cross-visitor emotional ecology			

03 The deployed experience

The whole thing runs in the browser — no install, no account. It reads as a wide desktop poster, then reflows to a single column on a phone: the colony scales to the screen, and you drag a creature with a mouse or a finger.



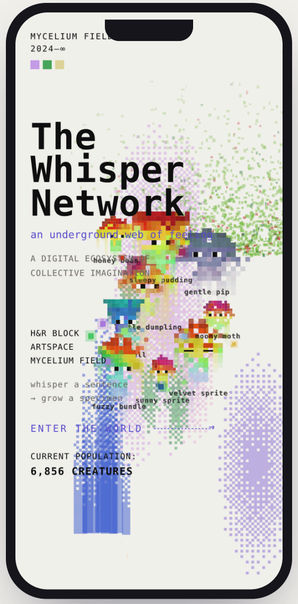
ON A DESKTOP

Anonymous whisper
a private sentence — no account, any language

Deterministic spore
the same sentence always grows the same creature

A shared colony
every visitor's feelings live in one field

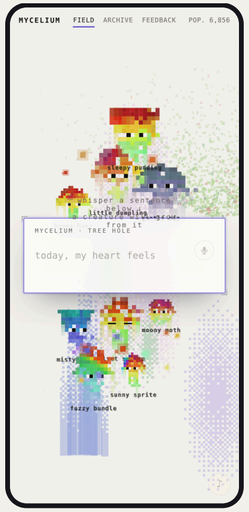
ON A PHONE



Connection as colour
bonds show as a cross-dye, not a chat

Care built in
mother-trees reach the ones left alone

Touch-native
drag a creature with a finger · EN / 中



TREE HOLE



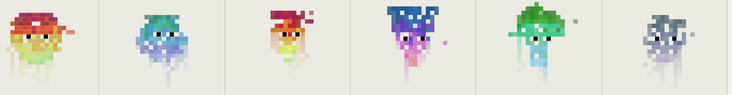
ARCHIVE



FEEDBACK

04 Emotion becomes form

Every whisper is read into one of six **emotion families**, each with its own base hue. Then the reading bends the form: intensity widens the palette from near-monochrome to rainbow, density sets how solid it is, agitation sets the jitter, and a second emotion blends in one accent. Same sentence, same creature — but no two feelings land the same.



tender	calm	curious	dreamy	companion	lonely
温柔	平静	好奇	梦幻	陪伴	孤独
H 24°	H 205°	H 32°	H 268°	H 158°	desat.



low
near-monochrome

mid
a few accents

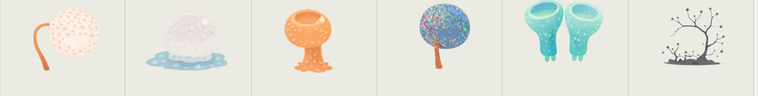
high
full rainbow

intensity → palette width **density** → solid vs. wispy **agitation** → jitter & drift

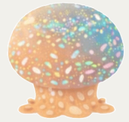
2nd emotion → one accent hue

05 How the creatures evolved

They didn't start as pixels. The first build shipped **six illustrated species** — soft watercolour fungi — plus fifteen hand-drawn hybrid forms. It was lovely, and it read wrong: with a fixed image library, two very different sentences could wear the same body. Emotion had become a label that picks a sticker.



radial	bubble	mushroom	glitter	cups	shrub
v1 · illustration	v1 · illustration	v1 · illustration	v1 · illustration	v1 · illustration	v1 · illustration
fixed sprite	fixed sprite	fixed sprite	fixed sprite	fixed sprite	fixed sprite



hybrid A×B
one of 15, drawn by hand



hybrid C×E
every pair pre-decided



hybrid D×F
finite by construction

WHAT CHANGED	WHAT SURVIVED	WHY IT MATTERED
The image library became a grammar. Three silhouette equations × six colour families, seeded by the sentence itself — one body per *sentence*, not per species. The fifteen hybrid PNGs became a live state machine.	The six-family taxonomy, the mushroom silhouettes — capped, round and bell trace straight back to these drawings — the gentle palette, and the hybrid idea.	With mascots, your feeling picked a sticker. With rules, your sentence grows a body only it can grow — determinism is what makes the creature *yours*.

v1 6 fixed sprites · 15 drawn hybrids **v2** deterministic mosaic engine · endless variation

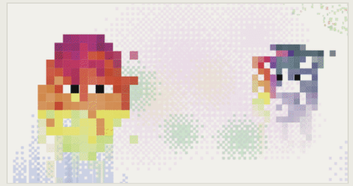
same soul six families · mushroom forms

06 How the colony behaves



Cross-dye encounter LIVE RENDER

Two compatible creatures drift together, take on each other's colour from the seam where they touch, trade matter tiles, then part and slowly fade back toward themselves.



Mother-tree support LIVE RENDER

An elder that has been on stage long enough grows larger, sways, and feeds a slow one-way stream to a creature that's been alone too long — which quietly warms toward the elder's palette.

Connection as colour

逐格染色 — a tendril in colour, not a line

When two compatible creatures meet, they dye each other starting from where they touch — strongest at the contact seam — and pass small matter tiles back and forth while bonded, breathing in sync.

Mother trees

母树 — care as an ecological rule

A creature that has been around long enough starts supporting an isolated one: it grows visibly larger, sways, flickers, and feeds it a slow one-way stream of matter. Nothing you press sets this off.

Marks that stay

永久杂交 — changed, not fused

A long enough encounter leaves a small, permanent trace of the other's palette. The creature keeps its own identity; the relationship just stays written on it.

07 The research behind it

RQ·main — how might anonymous emotional input become a perceivable, evolving collective emotional environment?

RQ2 — expression without public performance

RQ3 — AI as rule generator, not companion

RQ4 — emotion mapped to form, behaviour, relation

It draws on loneliness research (U.S. Surgeon General 2023; Baek et al. 2023), a critique of affective computing, and care-oriented design research (Hirsch). The full framing, methods, ethics and the sourced data sit in the research essay and case study — linked below.

08 From screen to substrate

The whisper doesn't have to stop at the screen. In the gallery installation, the colony's pooled emotion drives a living fungus, so the same anonymous sentence sets the growing conditions for a real culture. The software bridge is built and live; the physical cultivation is specced and staged.

whisper	/api/pulse	chamber	/api/bio
feeds the culture	aggregate climate	a living fungus	the loop closes
each sentence → one visible feeding pulse: mist + a light pulse in its emotion's hue	dominant mood → grow-light hue · intensity → mist · growth → cadence	sealed, HEPA-filtered · food-safe low-spore species	the fungus's biosignal feeds back — the projected colony breathes with it

DATA, NOT DECORATION

population, intensity, dominant family, agitation, loneliness — real aggregates from the live colony set real environment targets.

HONEST & SAFE

the projection shows only the emotion reading, never the sentence; biosafety is designed in for a weeks-long gallery run.

09 Read the full record

This page is the short version. The deeper documents carry the rest — the research it grew from, the full design-research write-up, the visual system that holds it together, and two printable artifacts.

01 · RESEARCH

Research & background

前期调研 · 背景故事

The loneliness research, the platform critique, and the sourced data behind the project — as a scrollable visual essay.

open the essay ▸

02 · CASE STUDY

Design-research case study

完整案例研究 · EN / 中

The full framing, method, decisions, ethics, and evaluation — the long-form account of how it was made and why.

open the case study ▸

03 · DESIGN SYSTEM

The visual language

设计系统板

Colour families, type, the deterministic spore grammar, and the motion rules — the board that keeps every screen consistent.

open the board ▸

04 · POSTER

Research poster, 7 sheets

横版研究海报 · PDF

The whole project on seven 2160×780 landscape sheets — background, evidence,

05 · INSTALLATION

Installation design manuscript

装置设计手稿 · PDF

Five drawing sheets for the gallery build — site plan, elevations, axon cutaway, and the

system, creatures, evaluation — in the site's own visual language.

download the PDF ▸

living-culture chamber that breathes with the colony.

download the PDF ▸

MYCELIUM · The Whisper Network

Concept · design · full-stack build — React · canvas 2D · Claude · Edge · Redis.

The same sentence always grows the same spore.

Every specimen on this page is rendered by the product's own engine — the same pixels the live colony grows.